MODULE 1 · LECTURE 26

AI in Precision Agriculture: Yield Mapping and Predictive Analysis

How combine yield monitors measure yield every second, why raw yield data must be cleaned, how points become maps and maps become management zones, and how machine learning predicts yield before harvest — with worked examples, a yield-mapping program and India's FASAL and YES-TECH systems.

Learning objectives

After this lecture you should be able to:

1. Explain precision agriculture and its data–map–decide–apply–evaluate cycle.
2. Describe the sensors of a combine yield monitor and how they are calibrated.
3. Calculate yield from monitor readings and correct it to a standard moisture content.
4. Explain the main errors in yield data and how cleaning and interpolation turn points into maps.
5. Explain how yield maps and other layers are used to delineate management zones.
6. Describe machine-learning yield prediction, its evaluation, and India's FASAL and YES-TECH systems.

Scope: Lecture 12 covered variable-rate applicators, Lecture 15 remote sensing and GIS, Lecture 11 regression and model evaluation, and Lectures 23–24 data, IoT and machine learning. This lecture uses them to measure, map and predict yield — the evaluation step of precision agriculture and the starting point for the next season's decisions.

1. Precision agriculture

Yield, soil and crop condition vary from place to place within a field and from season to season. **Precision agriculture** combines sensors, information systems, enhanced machinery and informed management to optimise production by accounting for this variability and uncertainty; adapting inputs site-specifically allows better use of resources while protecting the environment (Gebbers & Adamchuk, 2010).

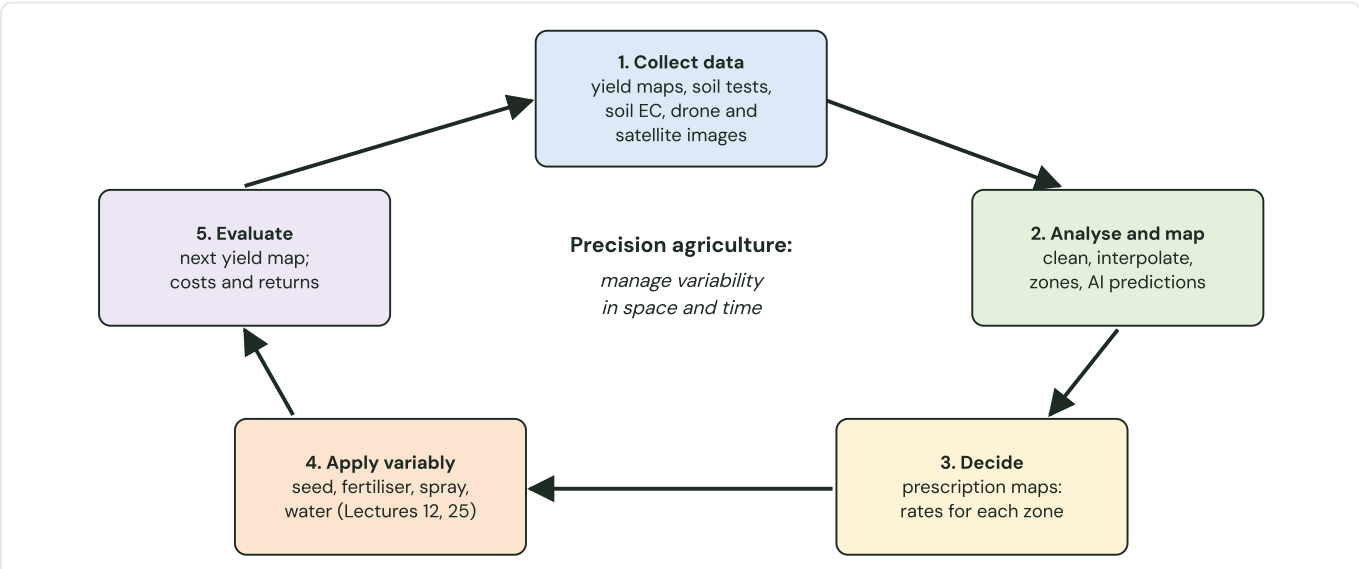


Figure 1. The precision agriculture cycle. Original diagram.

The yield map closes the cycle: it shows the result of this season's management and is the main layer for planning the next.

2. Yield monitors

A grain yield monitor on a combine harvester measures grain flow, grain moisture, area covered and location, and calculates yield automatically; with a differentially corrected GNSS receiver, every reading becomes a point on a map (Grisso et al., 2009).

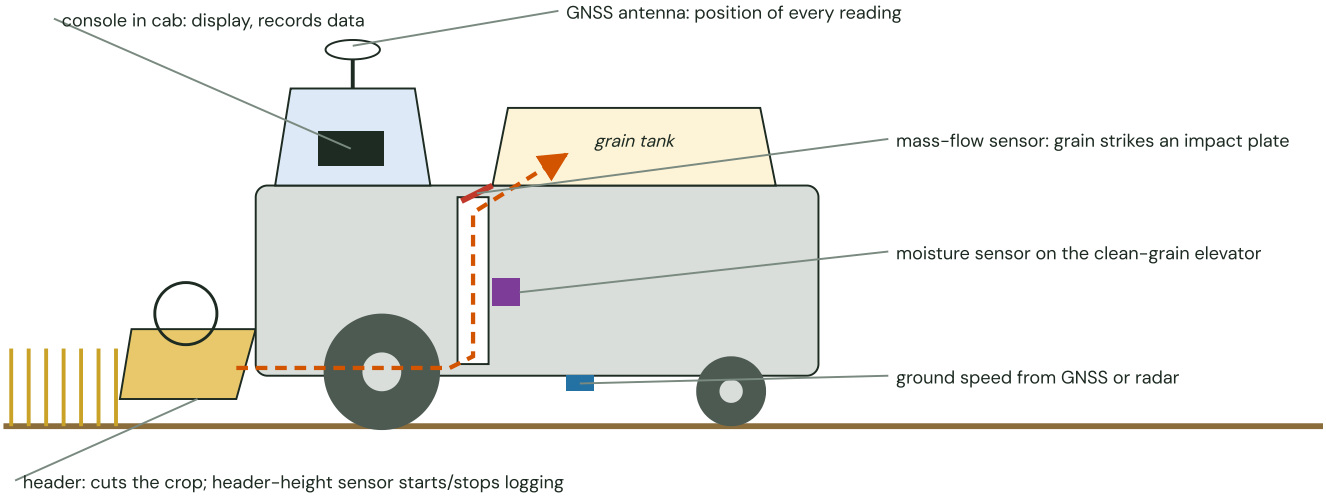


Figure 2. Sensors of a combine yield monitor. Original schematic, not to scale.

Component	What it does
Mass-flow (yield) sensor	Measures the flow of clean grain — most commonly the force of grain striking a plate at the top of the clean-grain elevator
Moisture sensor	Measures grain moisture in the clean-grain elevator or auger, so yield can be corrected to a standard moisture
Ground-speed sensor	Speed from GNSS, radar or the transmission; with header width gives the area harvested
Header-height sensor	Starts and stops data logging and area counting when the header is lowered or raised
GNSS receiver	Records the position of each reading
Console and data storage	Displays yield and moisture, stores readings every 1–3 seconds, holds calibration factors

Source of component descriptions: Grisso et al. (2009). In their example, a combine logging every 2 seconds collected about 200 yield readings per acre (about 500 per hectare) — more data than any other precision-agriculture tool.

Calculating yield

$$\text{Yield (t/ha)} = \text{mass flow (kg/s)} \times 10 \div [\text{speed (m/s)} \times \text{working width (m)}]$$

$$\text{Yield at 14\% moisture: } Y_{14} = Y \times (100 - M) \div (100 - 14)$$

Mass flow divided by the area harvested each second gives kg/m², and 1 kg/m² = 10 t/ha. Because wet grain weighs more, yield is converted to a standard moisture content, here 14% (wet basis), where M is the measured moisture in per cent; the dry matter stays the same.

> **Worked example 1 — from sensor reading to yield**
> Mass flow 8.0 kg/s of grain at 18% moisture; speed 5 km/h = 1.39 m/s; header width 6 m.
> Area per second = 1.39 × 6 = 8.33 m²/s → wet yield = 8.0 × 10 ÷ 8.33 = **9.60 t/ha**.
> At 14% moisture: 9.60 × (100 − 18) ÷ (100 − 14) = 9.60 × 82 ÷ 86 = **9.15 t/ha**.
> If the grain takes 12 s to travel from the header to the sensor, this reading belongs to crop cut 12 × 1.39 ≈ 16.7 m behind the combine.

Calibration

A yield monitor must be calibrated for each grain type at the start of harvest — for distance, header height, mass flow and moisture — usually by weighing several loads of grain and entering the weights, and recalibrated when moisture falls or crop conditions change (Grisso et al., 2009).

3. Errors in yield data

Raw yield maps contain many errors: researchers have reported that 10–50% of the readings in a field contain significant errors and should be removed (Sudduth & Drummond, 2007). The main causes are:

Error	Cause	Correction
Flow delay	Grain takes several seconds to pass from the header to the sensor — about 15 s in one example (Grisso et al., 2009) — so readings are logged at the wrong place	Shift each reading back by the delay
Filling and emptying	Low or zero readings when entering the crop, and grain still flowing after leaving it; the combine also smooths sudden changes	Remove the start and end of each pass
Narrow passes	Header not full (point rows, last pass), but full width is assumed	Record actual width or remove the pass
Surges and stoppages	Blockages, sudden speed changes	Remove outliers beyond set limits
Position errors	Poor GNSS signal near trees and buildings	Remove points off the field or out of line
Calibration drift	Moisture and crop changes during harvest	Recalibrate; check against weighed loads

Software such as Yield Editor applies such filters and lets the user check their effect on a map (Sudduth & Drummond, 2007).

4. From points to maps

Yield readings lie along the harvest passes, a few metres apart. To make a continuous map, values are estimated at the centre of each grid cell from nearby readings — **interpolation**. The simplest common method, **inverse distance weighting (IDW)**, goes back to Shepard (1968): nearer readings get more weight.

$$\hat{z}(x_0) = \Sigma\, w_i\, z_i \div \Sigma\, w_i \qquad w_i = 1 \div d_i^{\,p}\ (p = 2 \text{ is usual})$$

Here z_i are the measured yields within a search radius, d_i their distances from the cell centre and p the power. Kriging, a geostatistical method, chooses the weights from the spatial correlation of the data instead.

5. Worked example: building a yield map

The program simulates a combine with a 6 m header harvesting a 200 m × 120 m field at 5 km/h, logging one reading per second. The field has a fertile low area and a sandy knoll. The monitor has a 12-second flow delay, low readings while the flow builds up at the start of each pass, 6% random noise and occasional surges and stoppages. The program cleans the data, interpolates a 10 m grid by IDW and divides the field into zones.

Cleaning yield-monitor data, mapping by IDW and defining zones in Python (Lecture 8)

```
import math, random

random.seed(3)
LENGTH, WIDTH, HEADER = 200, 120, 6.0      # field (m) and header width (m)
SPEED, LAG = 5 / 3.6, 12                   # 5 km/h in m/s; grain takes 12 s to reach the sensor

def true_yield(x, y):                      # t/ha: a fertile low area and a sandy knoll
    return (6.0 + 2.0 * math.exp(-((x - 140) ** 2 + (y - 80) ** 2) / 2450)
                - 2.5 * math.exp(-((x - 50) ** 2 + (y - 30) ** 2) / 1250))

def x_at(p, second):                       # serpentine passes: odd passes run backwards
    return second * SPEED if p % 2 == 0 else LENGTH - second * SPEED

# 1. Simulate the yield monitor: one reading per second, logged at the combine's position
log = []                                   # (pass, second, x, y, reading in t/ha)
for p in range(int(WIDTH / HEADER)):
    y = HEADER * (p + 0.5)
    for k in range(int(LENGTH / SPEED) + LAG):
        if k < LAG:
            r = 0.0                        # grain has not yet reached the sensor
        else:
            r = true_yield(x_at(p, k - LAG), y) * random.gauss(1, 0.06)
            if k - LAG < 3:
                r *= (0.5, 0.7, 0.85)[k - LAG]    # flow still building up
            if random.random() < 0.01:
                r *= random.choice((0.2, 2.5))    # stoppage or surge
        log.append((p, k, x_at(p, k), y, r))

# 2. Clean: move each reading back to where its grain was cut, then drop bad points
clean, start_drop = [], 0
for p, k, x, y, r in log:
    cut = k - LAG                          # second at which this grain was cut
    if cut < 3:                            # combine filling or flow building up
        start_drop += 1
        continue
    clean.append((x_at(p, cut), y, r))
vals = [r for _, _, r in clean]
mean = sum(vals) / len(vals)
sd = (sum((v - mean) ** 2 for v in vals) / len(vals)) ** 0.5
kept = [(x, y, r) for x, y, r in clean if abs(r - mean) <= 3 * sd]

# 3. Map: inverse-distance-weighted yield at the centre of each 10 m x 10 m cell
def idw(points, cx, cy, radius=10):
    num = den = 0.0
    for x, y, r in points:
        d2 = (x - cx) ** 2 + (y - cy) ** 2
        if d2 <= radius ** 2:
            w = 1 / max(d2, 0.25)
            num += w * r
            den += w
    return num / den

cells = [(cx, cy) for cy in range(5, WIDTH, 10) for cx in range(5, LENGTH, 10)]
raw_pts = [(x, y, r) for _, _, x, y, r in log]
grid = {c: idw(kept, *c) for c in cells}
rmse = lambda pts: (sum((idw(pts, *c) - true_yield(*c)) ** 2 for c in cells) / len(cells)) ** 0.5
field_mean = sum(grid.values()) / len(grid)

print(f"Readings logged: {len(log)}; removed: {start_drop} at pass starts, "
      f"{len(clean) - len(kept)} outliers; kept: {len(kept)}")
print(f"Map error vs true yield (RMSE): raw {rmse(raw_pts):.2f} t/ha, cleaned {rmse(kept):.2f} t/ha")
```

```
print(f"Field mean yield: {field_mean:.2f} t/ha")
for name, lo, hi in (("Low", 0, 0.9), ("Medium", 0.9, 1.1), ("High", 1.1, 9)):
    zone = [v for v in grid.values() if lo * field_mean <= v < hi * field_mean]
    print(f"{name:6s} zone: {100 * len(zone) / len(grid):4.1f}% of field, mean {sum(zone) /
len(zone):.2f} t/ha")

Output:
Readings logged: 3120; removed: 300 at pass starts, 30 outliers; kept: 2790
Map error vs true yield (RMSE): raw 1.54 t/ha, cleaned 0.17 t/ha
Field mean yield: 6.19 t/ha
Low    zone: 22.5% of field, mean 4.75 t/ha
Medium zone: 49.2% of field, mean 6.19 t/ha
High   zone: 28.3% of field, mean 7.33 t/ha
```

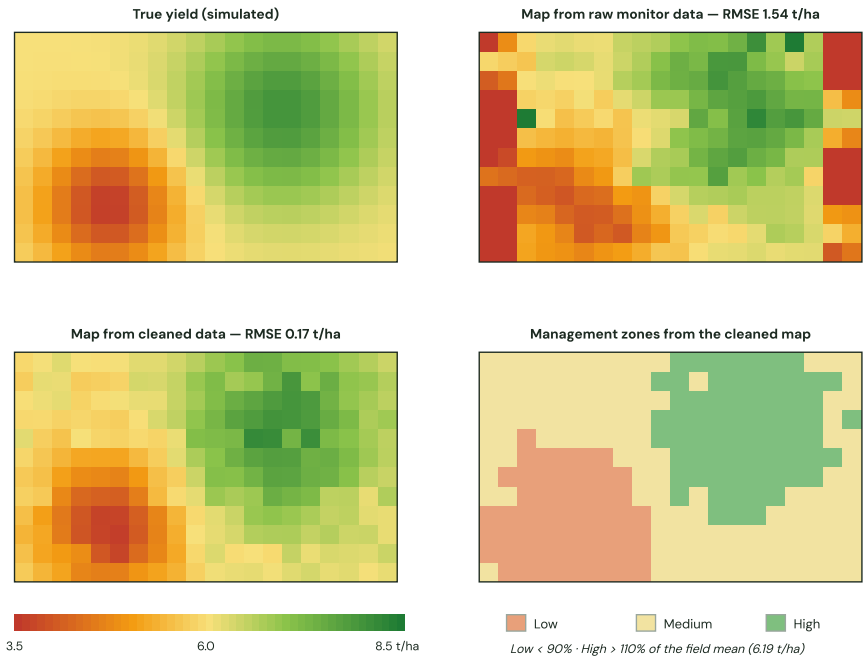


Figure 3. True yield, the map from raw monitor data, the map from cleaned data and the resulting management zones (each cell 10 m × 10 m). Original maps generated from the program's data.

- Cleaning removed 330 of 3,120 readings (11%) — within the 10–50% range reported for real fields.
- The raw map has an error (RMSE) of 1.54 t/ha against the true yield, mostly from the zero readings logged while the combine fills. Even without them, the uncorrected 12-second delay — 16.7 m along each pass, in opposite directions on alternate passes — leaves an error of 0.64 t/ha.
- Shifting readings back to where the grain was cut lowers the error to 0.31 t/ha; dropping start-of-pass points and outliers lowers it to **0.17 t/ha**.
- The cleaned map divides the field into a low zone (22.5% of the area, 4.75 t/ha), a medium zone (49.2%, 6.19 t/ha) and a high zone (28.3%, 7.33 t/ha).

The program was run and the output shown is its actual output; the error breakdown in the second and third points was checked by re-running the mapping step. The field, yields and monitor errors are simulated with illustrative values.

6. Management zones

A **management zone** is a part of a field with similar yield potential that can be managed alike — for example given one seed rate or fertiliser rate. Zones are found by **unsupervised clustering** (Lecture 24) of layers such as yield maps from several years, soil electrical conductivity, elevation, slope and satellite

images. Management Zone Analyst, for example, uses fuzzy c-means clustering and gives performance indices to help decide how many zones to create (Fridgen et al., 2004).

- Use several years of yield maps: a zone that is high in one year and low in the next is unstable and needs different treatment.
- Check zones in the field: soil samples, crop observations and the farmer's knowledge explain why a zone differs.
- Turn zones into **prescription maps** for variable-rate seeding, fertiliser or irrigation (Lecture 12), and judge the result with the next yield map.

7. Predictive analysis: forecasting yield

Predictive analysis uses past and current data to forecast what will happen — yield before harvest, disease risk (Lecture 25), water demand (Lecture 13) or prices (Lecture 14). Machine learning supports decisions on which crops to grow and what to do during the season. A systematic review of 50 machine-learning yield-prediction studies found that the most used inputs were temperature, rainfall and soil type, and the most applied algorithm was the artificial neural network; among 30 deep-learning studies, CNNs were used most, followed by LSTM and deep neural networks (van Klompenburg et al., 2020).

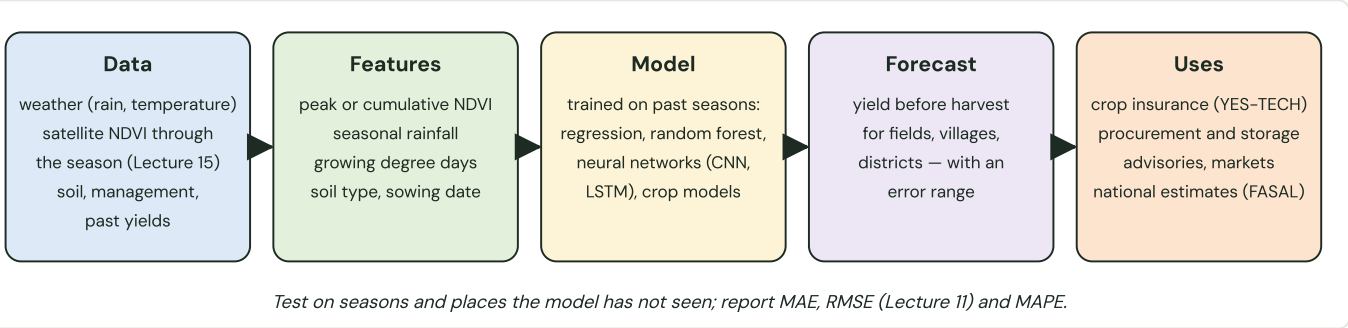


Figure 4. How a yield-prediction system works. Original diagram.

Statistical and machine-learning models learn the link between inputs and yield from past seasons; crop simulation models (Lecture 11) calculate crop growth from physiology. Both must be tested on seasons and places they were not built on.

Worked example 2 — testing a yield forecast
A model trained on 2016–2022 predicts village yield from peak NDVI: yield (t/ha) = -0.5 + 9.0 × NDVI (illustrative).
Test years: 2023, NDVI 0.71 → predicted 5.89, actual 6.20 (error -0.31); 2024, NDVI 0.62 → 5.08 vs 4.80 (+0.28); 2025, NDVI 0.68 → 5.62 vs 5.90 (-0.28).
Mean absolute error MAE = (0.31 + 0.28 + 0.28) ÷ 3 = **0.29 t/ha**.
Mean absolute percentage error MAPE = (0.31/6.20 + 0.28/4.80 + 0.28/5.90) ÷ 3 × 100 = (5.0 + 5.8 + 4.7) ÷ 3 ≈ **5.2%**.
A forecast is useful only if its error is small compared with the year-to-year change it must detect — here yields varied by 1.4 t/ha.

$$\mathrm{MAE} = \Sigma|P_i - O_i| \div n \qquad \mathrm{MAPE} = (100 \div n) \times \Sigma|P_i - O_i| \div O_i$$

Yield estimation in India

- **FASAL** (Forecasting Agricultural output using Space, Agro-meteorology and Land based observations): the Mahalanobis National Crop Forecast Centre (MNCFC) forecasts crop production at

district, state and national levels for major crops using optical and microwave remote sensing, with methods developed by ISRO's Space Applications Centre; yield estimates use semi-physical and crop simulation models with ISRO and IARI, and around 90,000 ground-truth surveys were made with the KrishiMapper app in kharif 2024–25 (MNCFC, n.d.).

- **YES-TECH** (Yield Estimation System based on Technology) under the Pradhan Mantri Fasal Bima Yojana moves crop-insurance yield estimation towards remote sensing: launched for paddy and wheat from kharif 2023 with a mandatory 30% weight for YES-TECH-derived yield, extended to soybean from kharif 2024 (Press Information Bureau [PIB], 2025), and raised to up to 50% in certain states (PIB, 2026).

8. Benefits and challenges

Benefits	Challenges
Shows where and how much yield varies; evaluates every management change	Raw yield data are full of errors; cleaning needs skill and software
Zones and prescription maps put inputs where they pay	Yield monitors need careful calibration; small plots are harvested in only a few passes, giving few readings per field
Forecasts support insurance, procurement, storage and trade decisions	Forecasts can fail in unusual seasons that differ from the training data
Remote-sensing estimates cover millions of small fields quickly (FASAL, YES-TECH)	Small, mixed fields and cloud cover make satellite estimates harder; ground truth is still needed
Records of yield history raise the value of farm data (Lecture 23)	Data ownership and sharing among farmers, service providers and insurers

9. Key takeaways

- Precision agriculture manages variability in a cycle: collect, analyse and map, decide, apply variably, evaluate.
- A yield monitor combines mass-flow, moisture, speed and header-height sensors with GNSS; it must be calibrated.
- Yield (t/ha) = mass flow × 10 ÷ (speed × width); correct to standard moisture with (100 − M) ÷ (100 − 14).
- Raw yield data contain 10–50% erroneous readings; flow delay, filling and outliers must be corrected before mapping.
- IDW interpolation turns points into maps; clustering of maps and soil layers gives management zones.
- ML yield prediction mostly uses weather and soil inputs with neural networks; forecasts must be tested on unseen seasons (MAE, RMSE, MAPE).
- India uses remote sensing for crop forecasts (FASAL) and for crop-insurance yields (YES-TECH, from kharif 2023).

10. Examination questions

Short-answer questions (2 marks)

Q1. Define precision agriculture.

Management that combines sensors, information systems, machinery and informed decisions to account for variability within fields.

Q2. Name four sensors of a combine yield monitor.

Mass-flow sensor, moisture sensor, ground-speed sensor, header-height sensor (plus GNSS receiver).

Q3. How does the common mass-flow sensor work?

It measures the force of grain striking an impact plate at the top of the clean-grain elevator.

Q4. Write the formula for yield from mass flow.

Yield (t/ha) = mass flow (kg/s) × 10 ÷ [speed (m/s) × width (m)].

Q5. Correct 9.60 t/ha at 18% moisture to 14%.

9.60 × 82 ÷ 86 = 9.15 t/ha.

Q6. What is the flow-delay error?

Grain takes several seconds to reach the sensor, so readings are logged ahead of where the grain was cut.

Q7. What share of raw yield readings may contain significant errors?

10–50% (Sudduth & Drummond, 2007).

Q8. What is inverse distance weighting?

Interpolation in which each nearby reading is weighted by $1/d^p$, so nearer readings count more.

Q9. What is a management zone?

A part of a field with similar yield potential that can be managed with one input rate.

Q10. What is YES-TECH?

Yield Estimation System based on Technology under PMFBY, using remote sensing to estimate crop yields for insurance.

Long-answer questions (5–10 marks) — answer outlines

Q1. Explain the precision agriculture cycle and the role of yield maps.

Definition; Figure 1; data, mapping, decision, variable-rate application, evaluation.

Q2. Describe the components and calibration of a combine yield monitor and calculate yield from a reading.

Figure 2; component table; calibration elements; worked example 1.

Q3. Discuss the errors in yield-monitor data and how they are corrected.

Error table; flow delay and filling; outliers; Yield Editor; program results and Figure 3.

Q4. Explain how yield maps are made and used to define management zones.

IDW formula; kriging; program; zone results; clustering (Fridgen et al., 2004); prescriptions.

Q5. Describe machine-learning yield prediction and how a forecast is evaluated.

Inputs and algorithms (van Klompenburg et al., 2020); Figure 4; worked example 2; MAE, RMSE, MAPE.

Q6. Explain the use of remote sensing for crop yield estimation in India.

FASAL and MNCFC; YES-TECH in PMFBY; ground truth; benefits and challenges.

11. Glossary

Term	Meaning
Calibration (yield monitor)	Matching sensor readings to weighed loads of grain
Flow delay	Time grain takes from the header to the yield sensor
IDW	Inverse distance weighting interpolation
Impact plate	Plate whose force from striking grain measures mass flow
Interpolation	Estimating values between measured points
Kriging	Geostatistical interpolation based on spatial correlation
MAE	Mean absolute error of predictions
Management zone	Field area with similar yield potential, managed alike
MAPE	Mean absolute percentage error
Mass flow	Mass of grain passing the sensor per second
Precision agriculture	Managing inputs according to variability in space and time
Predictive analysis	Using data and models to forecast future outcomes
Prescription map	Map of input rates for variable-rate application
Standard moisture	Reference grain moisture for reporting yield
Yield map	Map of crop yield across a field
YES-TECH	Yield Estimation System based on Technology (PMFBY)

References

Fridgen, J. J., Kitchen, N. R., Sudduth, K. A., Drummond, S. T., Wiebold, W. J., & Fraisse, C. W. (2004). Management Zone Analyst (MZA): Software for subfield management zone delineation. *Agronomy Journal, 96*(1), 100–108. https://doi.org/10.2134/agronj2004.1000

Gebbers, R., & Adamchuk, V. I. (2010). Precision agriculture and food security. *Science, 327*(5967), 828–831. https://doi.org/10.1126/science.1183899

Grisso, R., Alley, M., & McClellan, P. (2009). *Precision farming tools: Yield monitor* (Publication 442-502). Virginia Cooperative Extension. https://vtechworks.lib.vt.edu/bitstream/handle/10919/51375/442-502.pdf

Mahalanobis National Crop Forecast Centre. (n.d.). *FASAL*. Ministry of Agriculture and Farmers Welfare, Government of India. Retrieved November 9, 2026, from https://www.ncfc.gov.in/fasal.html

Press Information Bureau. (2025, July 25). *Implementation of PMFBY* [Press release]. Ministry of Agriculture & Farmers Welfare, Government of India. https://www.pib.gov.in/PressReleasePage.aspx?PRID=2148513

Press Information Bureau. (2026, August 29). *Pradhan Mantri Fasal Bima Yojana: Affordable crop insurance for every farmer* [Backgrounder]. Ministry of Agriculture and Farmers Welfare, Government of India. https://www.pib.gov.in/PressReleasePage.aspx?PRID=2304549

Shepard, D. (1968). A two-dimensional interpolation function for irregularly-spaced data. In *Proceedings of the 1968 23rd ACM National Conference* (pp. 517–524). Association for Computing Machinery. https://doi.org/10.1145/800186.810616

Sudduth, K. A., & Drummond, S. T. (2007). Yield Editor: Software for removing errors from crop yield maps. *Agronomy Journal, 99*(6), 1471–1482. https://doi.org/10.2134/agronj2006.0326

van Klompenburg, T., Kassahun, A., & Catal, C. (2020). Crop yield prediction using machine learning: A systematic literature review. *Computers and Electronics in Agriculture, 177*, 105709. https://doi.org/10.1016/j.compag.2020.105709

All diagrams, maps, examples and programs are original. Sensor readings, the NDVI model and the simulated field are illustrative; every calculation and the program output were checked by running them.

Practice MCQs

Q1. Precision agriculture aims mainly to

(a) use the same input rate everywhere
(b) manage inputs according to variability within fields
(c) replace all farm labour
(d) increase field size

Q2. The most common yield-monitor mass-flow sensor measures

(a) the colour of grain
(b) the force of grain striking a plate at the top of the clean-grain elevator
(c) the sound of the engine
(d) the weight of the whole combine

Q3. Which sensor starts and stops yield logging?

(a) Moisture sensor
(b) Header-height sensor
(c) Impact plate
(d) Thermal camera

Q4. A mass flow of 8.0 kg/s at 1.39 m/s with a 6 m header gives a wet yield of about

(a) 0.96 t/ha
(b) 6.0 t/ha
(c) 9.6 t/ha
(d) 96 t/ha

Q5. Converting 9.60 t/ha at 18% moisture to 14% moisture gives

(a) 8.40 t/ha
(b) 9.15 t/ha
(c) 10.07 t/ha
(d) 9.60 t/ha

Q6. Sudduth and Drummond (2007) report that raw yield data contain significant errors in about

(a) 1% of readings
(b) 10–50% of readings
(c) 80–90% of readings
(d) no readings

Q7. The flow-delay error means that a reading is logged

(a) behind the true position
(b) ahead of where the grain was cut
(c) at the field centre
(d) twice

Q8. In inverse distance weighting, each reading's weight is

(a) proportional to its distance
(b) $1 \div d^p$
(c) equal for all readings
(d) random

Q9. Management zones are usually found by

(a) supervised classification of labelled images
(b) unsupervised clustering of yield and soil layers
(c) manual counting of plants
(d) reinforcement learning

Q10. In the review by van Klompenburg et al. (2020), the most applied machine-learning algorithm for yield prediction was

(a) decision trees
(b) artificial neural networks
(c) k-means
(d) linear programming

Q11. MAPE stands for

(a) maximum absolute prediction error
(b) mean absolute percentage error
(c) mean average plot estimate
(d) model accuracy percentage estimate

Q12. YES-TECH under PMFBY was launched for paddy and wheat from

(a) Kharif 2016
(b) Kharif 2023
(c) Rabi 2019
(d) Kharif 2026

Answers

Q1 — (b) manage inputs according to variability within fields

Q2 — (b) the force of grain striking a plate at the top of the clean-grain elevator

Q3 — (b) Header-height sensor

Q4 — (c) 9.6 t/ha

Q5 — (b) 9.15 t/ha

Q6 — (b) 10–50% of readings

Q7 — (b) ahead of where the grain was cut

Q8 — (b) $1 \div d^p$

Q9 — (b) unsupervised clustering of yield and soil layers

Q10 — (b) artificial neural networks

Q11 — (b) mean absolute percentage error

Q12 — (b) Kharif 2023